⚙️ Configuration

Provider: `OPENAI`

Model: `gpt-4o-mini`

Max Budget Cap: `$0.10`

Timeout: `10.0s`

Preset Demo Queries

Select a sample query:

Custom Query

🔍 Autonomous AI Research Agent

Predixion AI Platform Engineering Internship Demonstration

Enter Research Question:

Compare LangGraph, CrewAI, and Microsoft AutoGen for building production-ready multi-agent systems. Evaluate architecture, orchestration, tool integration, memory management, scalability,

🚀 ...

Researching: **Compare LangGraph, CrewAI, and Microsoft AutoGen for building production-ready multi-agent systems. Evaluate architecture, orchestration, tool integration, memory management, scalability, error handling, and limitations.**

✓ ✅ Research Workflow Complete!

Executive Short Answer

There is limited comparative evidence available for LangGraph, CrewAI, and Microsoft AutoGen, particularly regarding Microsoft and AutoGen. CrewAI and LangGraph show promise in multi-agent orchestration and architecture, but comprehensive evaluations are hindered by insufficient data.

Confidence L...	Verified Sources	Execution Time	Tokens Used
LOW	3	44....	8,6...

↑ Est: $0.0...

Confidence Analysis: LOW (Score: 30.5/100 [Hard Safety Override: Target comparison entity 'Microsoft' is completely missing from retrieved evidence]): Limited

coverage (25%), 10 verified source(s) (0 authoritative, 5 deep page extractions).

<details>
<summary>📊 Evidence Coverage Matrix across Entities & Dimensions</summary>

Coverage Ratio: 25% | **Covered:** 2/20 dimensions

	LangGraph	CrewAI	Microsoft	AutoGen	Ev
Architecture	COVERED	COVERED	PARTIAL	PARTIAL	PA
Scalability	PARTIAL	MISSING	MISSING	MISSING	MI
Integrations	MISSING	PARTIAL	MISSING	MISSING	MI
Limitations	PARTIAL	MISSING	MISSING	MISSING	MI

</details>

💡 Key Findings

1. CrewAI utilizes a multi-agent architecture that coordinates specialized agents for tasks like meal planning and grocery shopping, demonstrating effective orchestration and integration with external APIs.

2. LangGraph is employed in various applications, including a multi-agent code reviewer, showcasing its capability in managing workflows and integrating with tools like ChromaDB and Google Gemini.

📚 Verified Sources Used

<details>
<summary>🔗 AgentHound [GENERAL_REPO]</summary>

URL: https://github.com/Telefonica/AgentHound

Relevance: Provides insights into the analysis of multi-agent architectures, including CrewAI and LangGraph, focusing on risk evaluation and capability paths.

</details>

<details>
<summary>🔗 Meal and Grocery Planner with CrewAI [GENERAL_REPO]</summary>

URL: https://github.com/sheraztariq22/Meal-and-Grocery-

Relevance: Demonstrates CrewAI's multi-agent orchestration in a production-ready application, detailing

its architecture and agent functionalities.

🔗 Multi-Agent Code Reviewer [GENERAL_REPO]

URL: https://github.com/Ranjan-10000/multi-agent-code-reviewer

Relevance: Illustrates the use of LangGraph in organizing multi-agent workflows for code analysis, highlighting its integration capabilities.

🏛️ Source Authority Breakdown

GENERAL_REPO (8)

- GitHub - Telefonica/AgentHound: BloodHound-style risk analysis for multi-agent AI architectures. Parses CrewAI, Dify, LangGraph, or generic YAML into a typed capability graph and detects reachable attack paths from untrusted input to sensitive data or external action, with live control simulation. · GitHub (https://github.com/Telefonica/AgentHound)
- GitHub - sheraztariq22/Meal-and-Grocery-: A production-ready intelligent meal planning and grocery shopping assistant powered by CrewAI, LangChain, and Google Gemini. This system uses a multi-agent architecture to coordinate specialized AI agents for meal research, shopping organization, budget analysis, and waste reduction. · GitHub (https://github.com/sheraztariq22/Meal-and-Grocery-)
- multi-agent-code-reviewer - Ranjan-10000/multi-agent-code-reviewer (https://github.com/Ranjan-10000/multi-agent-code-reviewer)
- retail-shopping-assistant - NVIDIA-AI-Blueprints/retail-shopping-assistant (https://github.com/NVIDIA-AI-Blueprints/retail-shopping-assistant)
- Multi-Agent-Customer-Support - ANI-IN/Multi-Agent-Customer-Support (https://github.com/ANI-IN/Multi-Agent-Customer-Support)

- ForgeFlow - JoelJohnsonThomas/ForgeFlow (https://github.com/JoelJohnsonThomas/ForgeFlow)
- GitHub - Ranjan-10000/multi-agent-code-reviewer: A multi-agent code review system that uses RAG, LangChain, LangGraph, Gemini, and ChromaDB to analyze GitHub repositories for project overview, security, architecture, and code quality. · GitHub (https://github.com/Ranjan-10000/multi-agent-code-reviewer)
- swati1024/torrents · GitHub (https://github.com/swati1024/torrents)

COMMUNITY_ARTICLE (2)

- Microsoft Flight Simulator X - Wikipedia (https://en.wikipedia.org/wiki/Microsoft_Flight_Simulator_X)
- History of Microsoft Flight Simulator (https://en.wikipedia.org/wiki/History_of_Microsoft_Flight_Simulator)

📌 Key Assumptions

- Information is current based on publicly available documentation and technical publications indexed up to present.
- Benchmark and performance assertions assume standard production hardware environments unless otherwise noted.

⚠️ Limitations

- Insufficient independent evidence gathered for entity: 'Microsoft'.
- Insufficient independent evidence gathered for entity: 'AutoGen'.
- Comparative evidence for evaluation dimension 'Scalability' is incomplete.
- Comparative evidence for evaluation dimension 'Integrations' is incomplete.

- Comparative evidence for evaluation dimension 'Limitations' is incomplete.

🎯 Suggested Next Steps

- Conduct further research to gather independent evaluations of Microsoft AutoGen and its capabilities in multi-agent systems.
- Explore additional case studies or applications utilizing LangGraph and CrewAI to better understand their performance in production environments.

🔬 Agent Orchestration Trace (Observability & Retrieval Rounds)

Retrieval Rounds: 2

Second-Hop Queries: ['LangGraph scalability official documentation guide', 'Microsoft scalability official documentation guide', 'AutoGen scalability official documentation guide', 'Evaluate scalability official documentation guide']


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    "Evaluate the architecture of LangGraph, CrewAI, and Microsoft AutoGen."
    1 :
    "Investigate orchestration capabilities of LangGraph, CrewAI, and Microsoft AutoGen."
    2 :
    "Analyze tool integration options for LangGraph, CrewAI, and Microsoft AutoGen."
    3 :
    "Examine memory management strategies in LangGraph, CrewAI, and Microsoft AutoGen."
```

```
    4 : "Assess scalability features of LangGraph, CrewAI, and Microsoft AutoGen."
    5 : "Investigate error handling mechanisms in LangGraph, CrewAI, and Microsoft AutoGen."
    6 : "Identify limitations of LangGraph, CrewAI, and Microsoft AutoGen."
]
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]
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    1 : "CrewAI architecture analysis"
    2 : "Microsoft AutoGen architecture details"
    3 : "Microsoft AutoGen orchestration methods"
    4 : "LangGraph orchestration features"
    5 : "LangGraph scalability official documentation guide"
    6 : "Microsoft scalability official documentation guide"
    7 : "AutoGen scalability official documentation guide"
    8 : "Evaluate scalability official documentation guide"
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```
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  4 : "LangGraph orchestration features"
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Raw Validated JSON Output (Pydantic Schema)

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      "is_official": false
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    "Comparative evidence for evaluation dime
    "Comparative evidence for evaluation dime
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    ],
    [
      "Microsoft",
      "Integrations"
```

```
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```
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      "Multi-Agent-Customer-Support - ANI-IN/
      "ForgeFlow - JoelJohnsonThomas/ForgeFlo
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      "History of Microsoft Flight Simulator
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```

```
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